

2018-Ph-H1-Q5

Section: Electricity

Topic: Monitoring and Measuring a.c.

An alternating voltage has a peak value of 10 V.
What is the r.m.s. value?

Solution:

$$V_{rms} = \frac{V_{peak}}{\sqrt{2}} = \frac{10}{\sqrt{2}} \approx 7.07 \text{ V}$$

Rounded: 7.1 V

Answer: 7.1 V

Revision Tip:

The r.m.s. value is always lower than peak. Use $V_{rms} = \frac{V_{peak}}{\sqrt{2}}$ or vice versa.